

CASE STUDY

BLOCKCHAIN IN THE DIAMOND AND PRECIOUS METALS INDUSTRY

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INTRODUCTION

The diamond and precious metals industry faces several challenges such as lack of transparency, fraud, unethical sourcing, and complex supply chains. Diamonds and metals like gold pass through many intermediaries before reaching the final consumer. This makes it difficult to verify authenticity and origin. Blockchain technology offers a solution by providing a secure, transparent, and immutable ledger that records every transaction in the supply chain. This improves trust between miners, suppliers, retailers, and customers.

1

The Supply Chain

The diamond and precious metals industry involves many intermediaries such as miners, traders, wholesalers, and retailers. This complex supply chain makes it difficult to trace the origin and journey of diamonds and metals, reducing transparency and consumer trust.

2

Sourcing And Trading

Due to poor traceability and fragmented record-keeping, the industry is vulnerable to fraud, counterfeit products, and unethical sourcing practices such as conflict diamonds and illegally mined metals.

3

Retailers And Consumers

Blockchain technology provides a secure and immutable ledger that records every transaction in the supply chain, enabling stakeholders to verify authenticity, track origin, and ensure ethical sourcing of diamonds and precious metals.

Rampant smuggling of diamonds in State

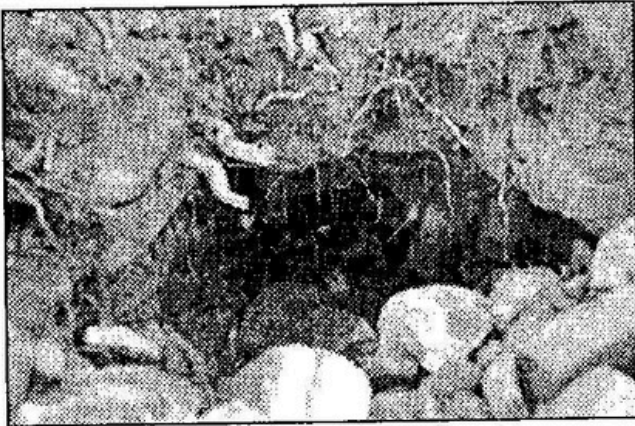
EJAZ Kaiser
Raipur, November 2

CLOSE ON the heels of the first ever diamond discovered by a poverty stricken tribal Barnu, in his agriculture land during the early nineties in Chhattisgarh, an unholy nexus also emerged in which large scale illegal prospecting and smuggling of raw gemstones by scores of locals and even outsiders was detected.

As soon as the news spread that this tribal State had rich deposits of diamonds, there was a mad rush by unscrupulous elements from within and outside the State to get rich quickly by frantic digging for the precious stones in the Deobhog and Mainpur

deep mining activities on diamonds owing to heavy investment involved in it.

While the State government has pitched in the Special Armed Force (SAF) personnel to track down smugglers and halt their activities in the prospective kimberlite rock areas, many criminals continue to operate unchecked. Kantibhai Patel who came down from Gujarat to Mainpur village with 'diamonds in his eyes', along with his friends was arrested on charges of illegal smuggling. Similarly, the law-enforcing agencies also nabbed quite a few businessmen from Maharashtra and Orissa who were camping in the State to illegally prospect for the precious stones.



The root of a tree dug in search of diamonds.

PHOTO/HT

Kamar tribals dig roots for diamonds

THE KAMAR tribal settlement near Deobhog subsists on 'Kandmool' (edible rootbeat) of plants. A Kamar member incidentally detected a precious diamond in the roots of

tribals who were encouraged for illegally tracing diamonds in return for insignificant amounts or exchanging clothes and other household goods.

It may be noted, that the name of Amit, son of former chief minister Ajit Jogi allegedly figured in one such case. It was disclosed in a court here that the flamboyant Amit was allegedly present when such illegal mining was being done. The police officer, who registered the case of illegal mining, however, informed the court that he could not take any action or even interrogate Amit as he was the son of the chief minister.

Taking strong exception over the tampered final investigation report present-

Barnu's cruel tryst with diamonds

EJAZ Kaiser
Raipur, October 21

FATE HAD played a cruel hand on him. Barnu, a tribal who is credited with finding the first ever diamond in his agricultural land in Chhattisgarh (then in Madhya Pradesh) in the early nineties died unsung a few months back mired in deep poverty. He could not even live long enough to secure the compensation for his land taken away by the State government. Barnu's diamond find heralded the discovery of this precious gem's raw form in this tribal State.

There are other tribals like Barnu who lead a poverty stricken life, blissfully unaware that the ground beneath their parched feet is probably studded with potential diamonds. The village folk of Paily Khan have little idea of how their lives might change. They have diamonds in their eyes and wait impatiently for the earth to throw up the diamonds after the government okays mining by the multi-nationals.

It was actually Barnu's find, that led mining inspectors to ex-



PHOTO/HT

Barnu's hut in Paily Khan village. (inset) Barnu.

Bhopal during the undivided Madhya Pradesh days to seek compensation promised to him, but to no avail.

Barnu's neighbours further informed that he was not even given alternative land to eke out his livelihood.

When contacted, the con-

Barnu leaves behind three sons and three daughters. They are still unable to fathom the reason as to why their late father was denied compensation and that too when the old man had lost his vision at the fag end of his life. It is ironic that Barnu's dependents continue to live in

CM- who? Yes, MLA had come for votes

THE POOREST among the community, the Bhunjia tribals residing in Paily Khan and Jangda villages, about 150 km from Raipur, are sitting on the richest land in Chhattisgarh. Barnu too was a Bhunjia. They are not only poor but ignorant also. These tribals have no idea about who is a chief minister. However, they could recollect that an MLA had come to their area for votes. The primary school facility in the area is just on name. Class V pass out Onkaresh does not know how to write his name, his father Shanturam said.

cense by the then Digvijay Singh government for diamond survey, prospecting and mining in the Paily Khan and Jangda regions.

Barnu's agricultural land falls in Paily Khan. Now Special Armed Force (SAF) personnel, who are deployed to prevent any smuggling or illegal digging of

PROBLEMS IN THE TRADITIONAL SYSTEM

TIMES IMPACT

Drone survey flags massive illegal mining in Aravallis near Bhilwara

Ajay Singh Upgani
@timesofindia.com

Jalpur: The Aravalli range near Bhilwara is being systematically damaged by large-scale illegal mining, with a drone-based volumetric survey exposing extensive excavation far beyond legally sanctioned limits.

The survey reveals a major mismatch between officially sanctioned extraction

MINING BEYOND LIMITS

- Total area mined | 64 hectares
- Legally sanctioned area | 11 hectares
- Drone-surveyed area | 28 hectares
- Illegal mining area identified | 41 hectares
- Total extraction (estimated) | 1.2 crore tonnes
- Extraction as per records | 62.76 lakh tonnes
- Estimated illegal extraction | 57 lakh tonnes
- Penalty per tonne | ₹400
- Estimated penalty | ₹254.5 crore



The Times of India
Dec 17, 2025

FINANCIAL TIMES

Amazon gold rush: Brazil grapples with illegal mining in the rainforest



Politicians paint mineral sand gray

By SHAFI RAHMAN

Alappuzha, Sept 22: The once full-throated political campaign against mining in the district's coastal region is wearing an injured look as the campaign for the on-coming local body elections in the state ends on Wednesday. The local politicians from both the UDF and LDF may have mixed feelings about the controversial mining debate, but the mining lobby has never missed a beat while appropriating tools and ideas to meet its ultimate wish - ensuring victory of politicians who would mould a panchayat resolution by okaying mining in the area.

Except for a few die-hard anti-mining leaders, many candidates and parties have refused to make anti-mining an election plank. Anti-mining diatribes have failed to find mention in the manifestoes of local candidates except in two wards of Arattupuzha panchayat.

Anti-mining campaigners suspect the credentials of some candidates who openly support mining. The mining companies and lobbies have been accused of lavishly funding candidates who protect their interests. "The local leadership of the CPM is sharply divided over the issue. The companies are making full use of the opportunity," says a CPM activist.

For mining companies, a resolution okaying mining is vital for further progress of the project, temporarily frozen by the state Government. A DYFI state-level leader has been accused of openly supporting a candidate from RSP(B), C.H. Sally, contesting from Thrikkunnapuzha to ensure the defeat of her CPI-leader opponent who is against mining.

But candidates from tsunami-hit wards five, six and seven of Arattupuzha have stepped up campaign against mineral sand mining. C.V. Sanyal, the CPM leader contesting from the area, known as a staunch anti-mining campaigner, is basing his campaign on the issue. "It is an issue closely identified with people of the area. I am sure my credential as an anti-mining campaigner will win me votes," says Sanyal.

DIC(K) leader K.K. Usman, seen widely as a supporter of opening up area of mining, has the backing of the LDF as he contest from Thrikkunnapuzha. In Arattupuzha and Thrikkunnapuzha panchayats, many candidates have refused to open up their mind, when Solidarity activists on Tuesday approached them to divulge their stand as a part of its anti-mining



CPM Candidate C.V. Sanyal, an anti-mining faithful, ensures the vote of Thekkathil Krishnan, a local fisherman, on the 'blessed' sands of Arattupuzha beach in Alappuzha.

Many Left leaders are refusing to buy the argument and mostly grumble the poor development of the area. Sluggish rehabilitation measures for tsunami-affected families have also turned the ire on

campaign. "Many of the candidates publicly oppose the mining but take their orders from the mining lobby," says M. Shameer, unit vice-president of Solidarity.

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the Government. Many of the families living in the makeshift camps are waiting for the Government to fulfill its promises.

Villagers also accuse the authorities of slow-peddling on a project to set up sea wall to prevent sea erosion in the area. Angry sea has already swallowed many dwellings in the area. "The idea is to scare away people from the area and to make village a deserted haven for mining companies," said C. Rajendran, a local fisherman.

Though the politicians scurry in different direction, the common thread between residents of the doomed coastal hamlets is still their unrelenting fight for the sand under their feet.

(The article has been written under the aegis of Centre for Science and Environment Media Fellowship)

PROBLEMS IN THE TRADITIONAL SYSTEM

1 Lack of Transparency

Consumers often have limited information about the origin of diamonds or precious metals they purchase. Because the supply chain involves many intermediaries, it becomes difficult to verify whether a diamond or gold product was sourced ethically and legally.



2. Conflict Diamonds (Blood Diamonds)

Conflict diamonds are diamonds mined in war zones and sold to finance armed conflict against governments. These diamonds may enter the global market through illegal channels and become mixed with legitimate supplies.



3. Fraud and Counterfeiting

The industry also faces issues related to fake diamonds and misrepresented precious metals. Some sellers may falsely label synthetic diamonds as natural or misrepresent the purity of gold and other metals.



4. Complex Supply Chain

The journey of diamonds and precious metals typically passes through multiple stages, including miner, trader, cutter, polisher, distributor, retailer, and finally the customer. Each stage involves separate transactions and ownership transfers.



5. Poor Record Keeping

In many cases, transaction records are maintained through paper documents or centralized databases. These systems are vulnerable to errors, data loss, or intentional manipulation.



BLOCKCHAIN BASED SOLUTION



Creation of a tamper-proof digital record for each diamond or metal asset



Recording of supply chain transactions on the blockchain



Tracking the origin of the diamond or precious metal



Recording ownership details at every stage



Maintaining processing and manufacturing history



Storing certification and authenticity information



Recording transportation and logistics details



Immutable and unchangeable transaction records on the blockchain

ARCHITECTURAL DIAGRAM

Step 2: Registration

Details recorded:
Mine location,
Weight, Quality, Date
of extraction

Step 1: Mining

When a diamond or
gold is mined, a unique
digital identity is
created on the
blockchain.

Step 3: Processing

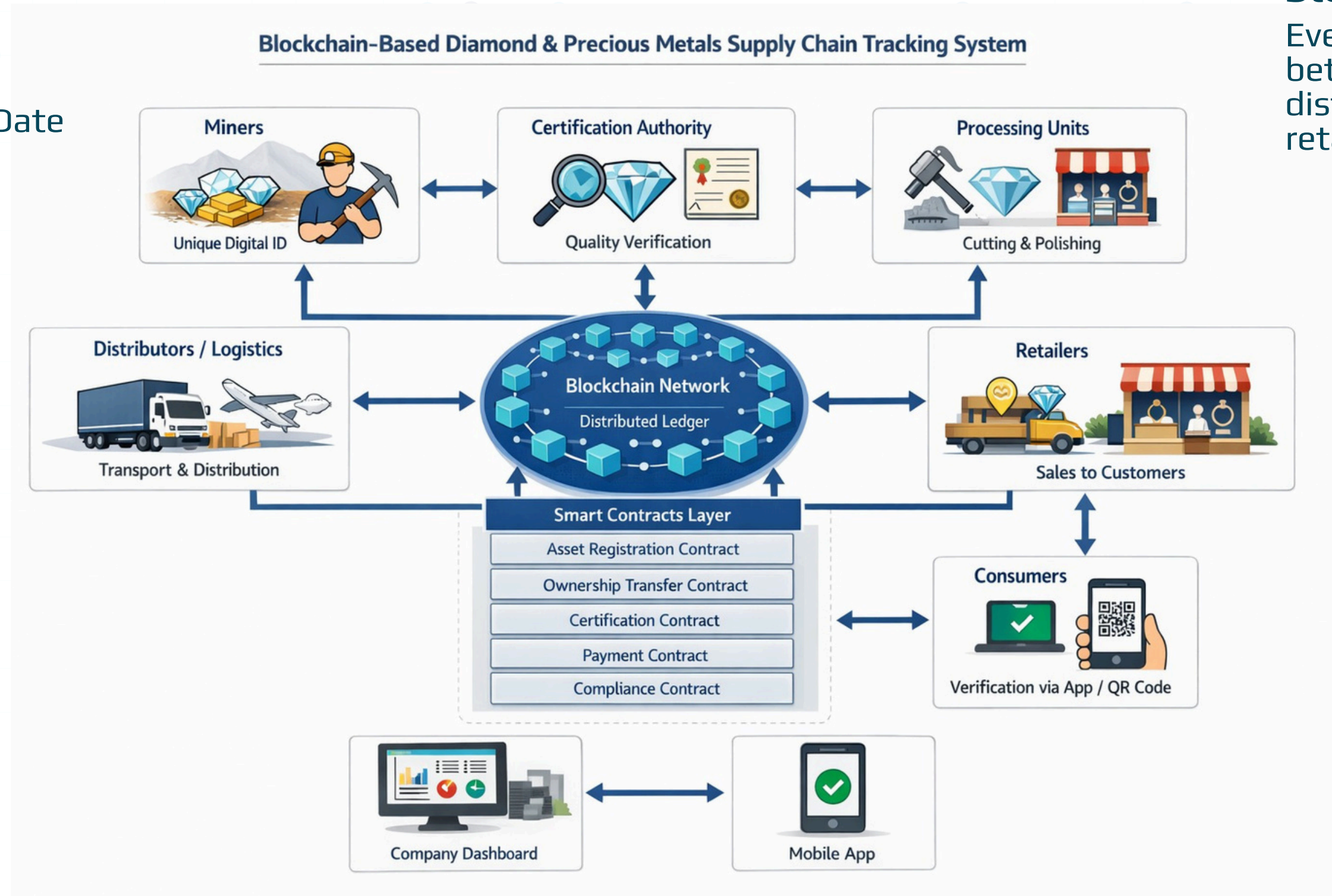
When diamonds are cut and polished, the information
is added to the blockchain.

Step 4: Distribution

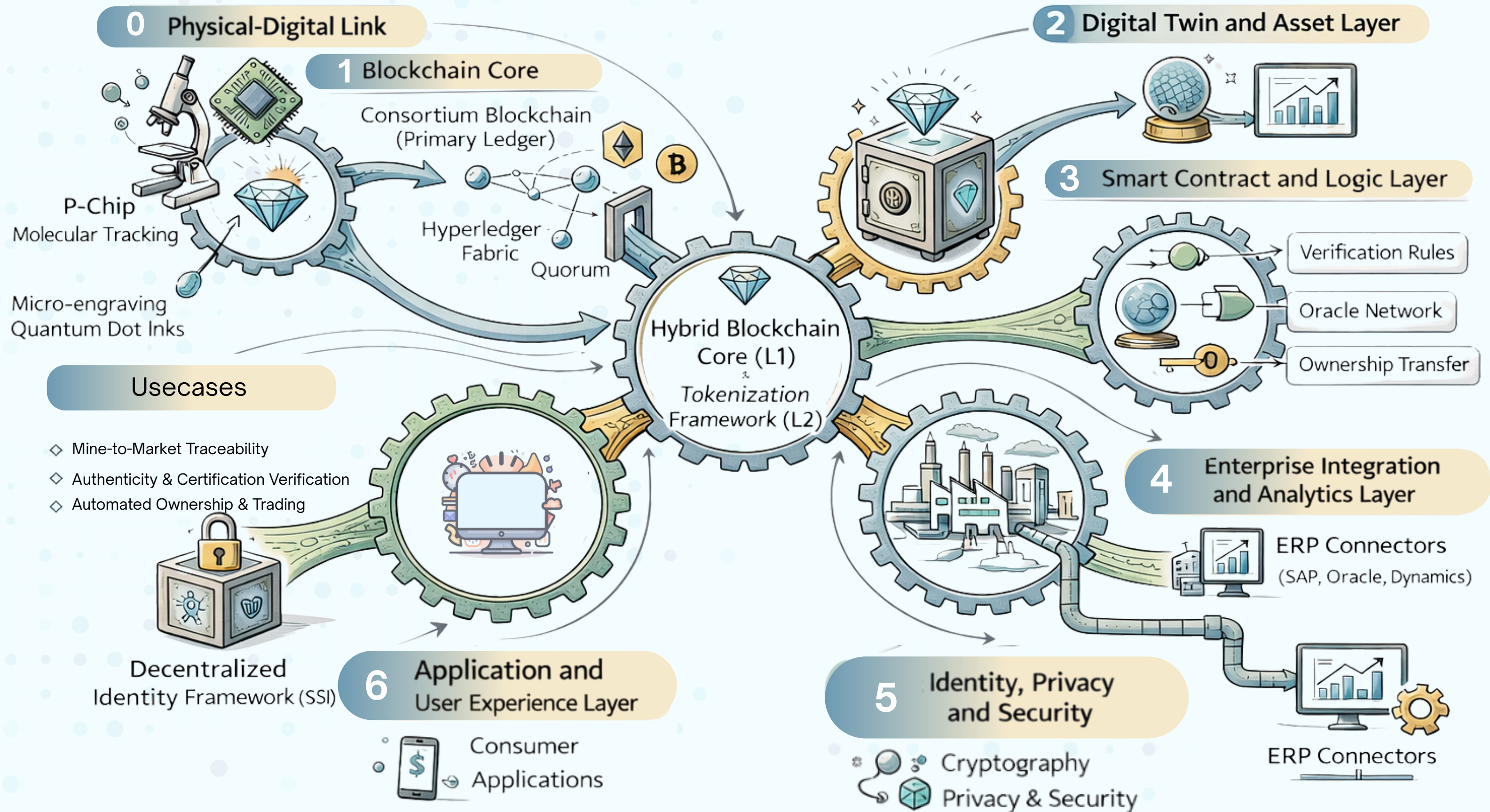
Every transfer
between suppliers,
distributors, and
retailers is recorded.

Step 5: Retail Sale

Consumers can verify
the full history of the
diamond or metal
using a QR code or
digital certificate.



REAL WORLD PARALLEL



SMART CONTRACTS THAT CAN BE APPLIED

Asset Registration Smart Contract

1. Registers newly mined diamonds or precious metals on the blockchain.
2. Functions
3. Assign unique asset ID
4. Store mine location
5. Record weight and quality

Ownership Transfer Smart Contract

1. Automatically updates ownership when a diamond moves from miner → trader → retailer.
2. Functions
3. Transfer ownership
4. Record transaction timestamp
5. Update supply chain records

Certification Smart Contract

1. Automatically updates ownership when a diamond moves from miner → trader → retailer.
2. Functions
3. Transfer ownership
4. Record transaction timestamp
5. Update supply chain records

Payment Smart Contract

1. Automates payments between parties once conditions are fulfilled.
2. Functions
3. Escrow payments
4. Release funds after delivery confirmation

Compliance Smart Contract

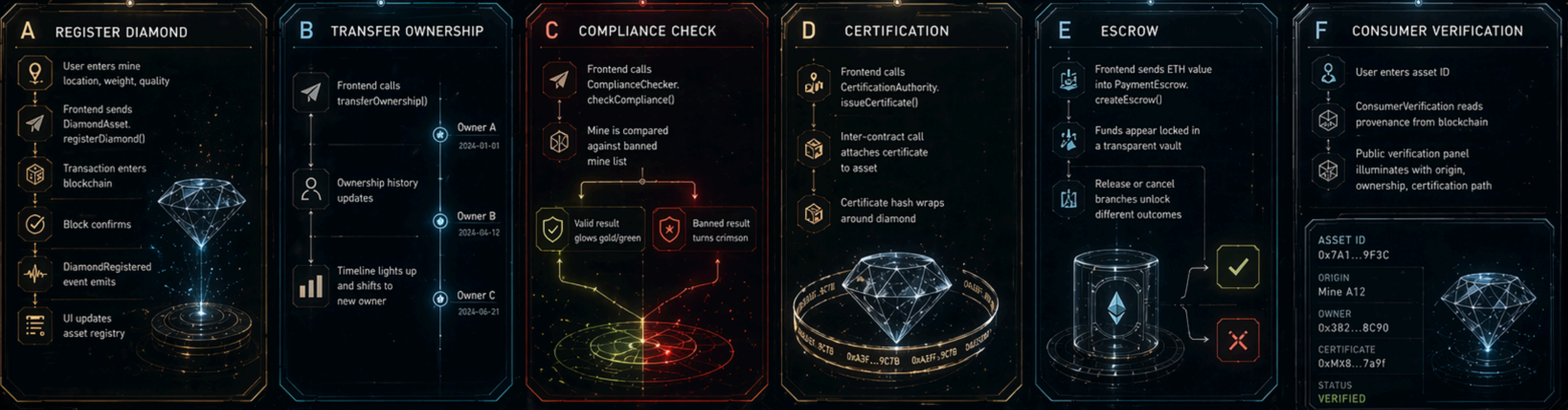
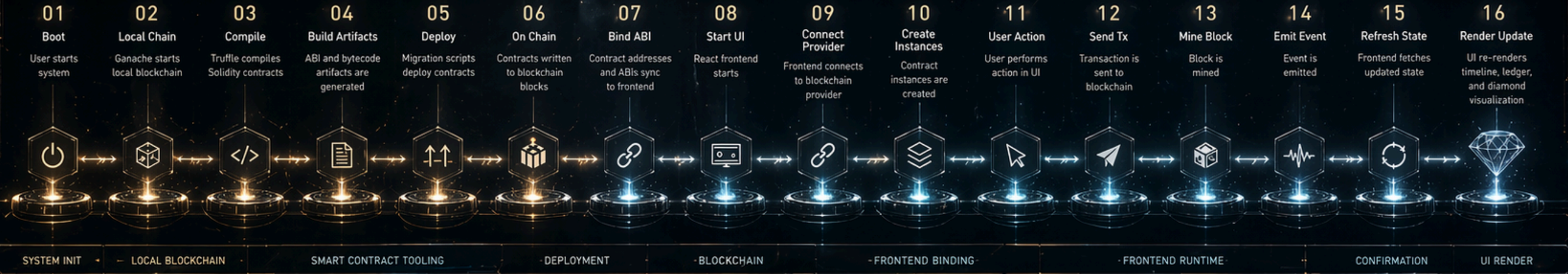
1. Ensures diamonds follow ethical sourcing regulations.
2. Functions
3. Verify mine legitimacy
4. Check regulatory compliance
5. Prevent trade of conflict diamonds

Consumer Verification Smart Contract

1. Allows customers to verify the authenticity and history of the diamond.
2. Functions
3. Provide supply chain history
4. Show certification details
5. Confirm ownership record

EXECUTION FLOW

SUPPLY CHAIN DAPP



Compile. Deploy. Connect. Execute. Confirm. Render.

SMART CONTRACT LIFECYCLE IN REMIX IDE

FROM SOURCE TO ON-CHAIN EXECUTION

01 COMPILE

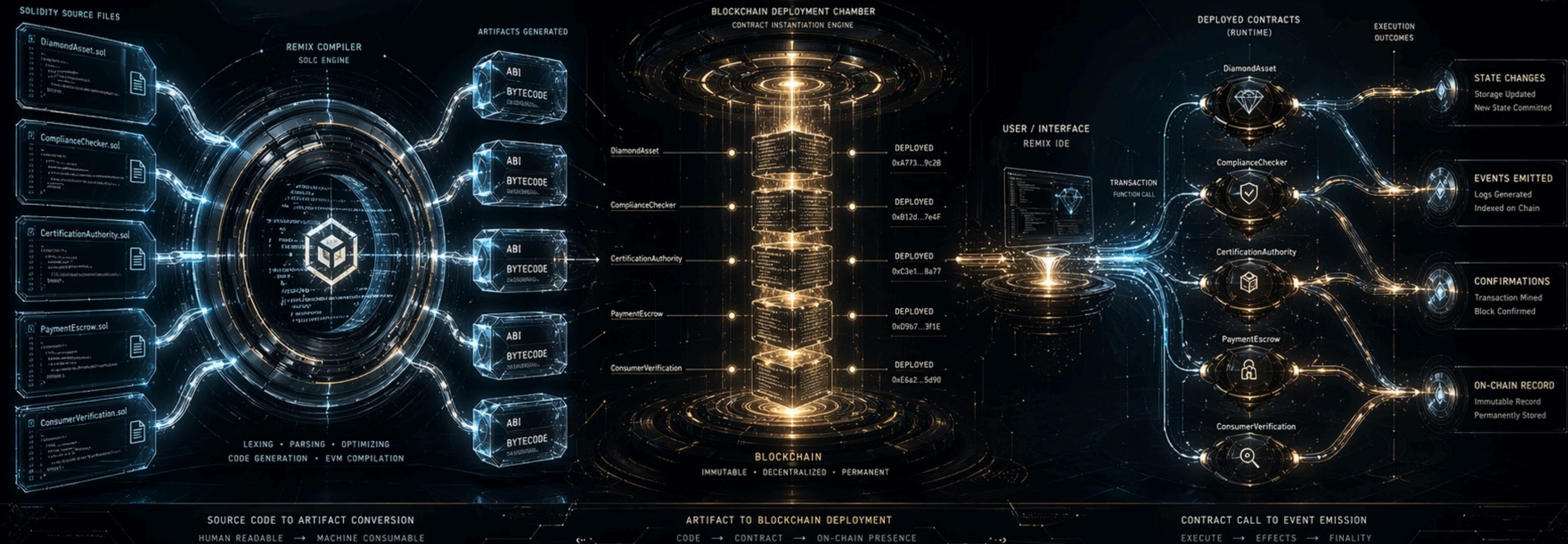
SOURCE CODE → ARTIFACTS

02 DEPLOY

ARTIFACTS → BLOCKCHAIN

03 TRANSACT

EXECUTE → EVENTS → STATE UPDATES



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THANK YOU